Ch 09: Serializing Objects

Application Program Development

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Serializing Objects (1 of 3)

Object Serialization

Serializing an object is the process of converting the object to a stream of bytes that can be saved to a file for later retrieval. In Python, object serialization is called pickling.

- **Serialize an object**: convert the object to a stream of bytes that can easily be stored in a file
- **Pickling**: Python name/method for serializing objects to a file

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Serializing Objects (2 of 3)

- To pickle an object
  - Import the `pickle` module
  - Open a file for binary writing
  - Call the `pickle.dump()` function
  - Close the file
- You can pickle multiple objects to one file prior to closing the file

```
import pickle
phonebook = {'Chris': '555-1111',
             'Katie': '555-2222',
             'Joanne': '555-3333'}
output_file = open('phonebook.dat', 'wb')
pickle.dump(phonebook, output_file)
output_file.close()
```



Serializing Objects (3 of 3)

- **Unpickling**: retrieving pickled objects back from files
- To unpickle an object:
  - Import the `pickle` module
  - Open an existing file with pickled data for reading in binary mode
  - Call the `pickle.load()` function
  - Close the file
- You can unpickle multiple objects from the file

```
import pickle
input_file = open('phonebook.dat', 'rb')
phonebook = pickle.load(input_file)
input_file.close()
```



Summary

- Serializing an object is the process of converting an object to a stream of bytes that can be saved to a file
- You can serialize multiple object to a single file (not shown)
- The files should be opened in `binary` mode since the data is a stream of bytes
- `pickle.dump()` will serialize and save a single object to a file
- `pickle.load()` will do the reverse and unserialize a byte stream from a file back to an object



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References



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